

title: "COMMUNITY TECHNICAL BRIEFING & ACTION DOSSIER" subtitle: "Glen Miller Sonoco Data Center: Open House Inquiry Sheet & Evidence Ledger" date: "October 6, 2026" author: "Quinte West Community Infrastructure & Environmental Working Group"

Facility: 920 Trenton-Frankford Rd / 5 Bernard Long Rd, Glen Miller

Proponents: FirstBlock, All Season Fencing (2874059 Ontario Inc.), Trinity Smart Farms

Public Notice: Verified evidentiary records, technical disclosures, and municipal filings compiled ahead of City Council's consideration of Interim Control By-law 26-131. Direct citations reflect FirstBlock's October 2, 2026 submission and regulatory records.

Metric / Parameter	Disclosed Figure	Underlying Technical Risk	Wiki Source Reference
Grid Power Allocation	18 MW vs. 33 MW	Phase 2 load hidden from public; unallocated transformer capacity	[[entities/hydro-one.md]] [[concepts/power-scale.md]]
Acoustic Low-Frequency	100.3 dB @ 125 Hz	Wall-penetrating hum omitted by dBA weighting; pollinator masking	[[raw/articles/04_acoustics_dba_dbc.md]] [[raw/articles/datacenterally-com-blog-why-dba-is-not-the-whole-sound-story.md]]
Vertical Survey Datum	CGVD28:78 vs 2013	Proponent unaligned comparison stamps LTC Sheet 112 as 'Faulty'	[[concepts/floodplain-mapping.md]] [[raw/articles/floodplain-mapping-cgvd28-cgvd2013.md]]
Chemical River Risk	1.68 kg O ₂ / L BOD	Closed-loop propylene glycol rupture causes river hypoxia at intakes	[[concepts/trent-river-drinking-water-intakes.md]] [[raw/articles/trent_river_intakes_and_ipz_source.md]]

1. POWER GRID SCALING & RATEPAYER PROTECTIONS

- **Grid Interdependency & Hidden Phase 2:** Hydro One's March 3, 2026 capacity allocation letter to 2874059 Ontario Inc. (All Season Fencing) assigns 18,000 kW out of the Sydney TS-M1 feeder with a strict Assigned Capacity Date of **January 2027**[cite: 1]. However, Page 7 of FirstBlock's October 2, 2026 Council filing discloses a **33 MW total draw** (16 MW in Phase 1 and 17 MW in Phase 2)[cite: 1].
- **On-Site Generation Asset:** The adjacent co-located parcel at 5 Bernard Long Road holds an active Ontario Energy Board generation licence (**EG-2010-0168**, valid until June 17, 2030) authorizing a **7.5 MW** generation facility[cite: 1].
- **Ratepayer Cost Exposure:** Substation expansions (\$35M–\$60M) and 230 kV line upgrades (\$2.5M–\$4.5M/km) risk being transferred to Elexicon and Hydro One Class B residential rate bases under IESO cost-allocation structures if dedicated Connection and Cost Recovery Agreements are not executed[cite: 1, 2].
- **Wiki & Archive Evidence:** [[entities/hydro-one.md]] | [[concepts/power-scale.md]] | [[raw/articles/quintenews-com-2022-01-25-power-outages-what-happened-html.md]] [cite: 1, 4]

DIRECT QUESTION FOR PROPONENTS:

"Page 7 of your October 2 Council submission details a Phase 2 expansion to 33 MW total draw—where will the additional 15 MW originate once Sydney TS-M1 is maxed out, and will FirstBlock sign a binding Connection and Cost Recovery Agreement backed by an irrevocable letter of credit so zero upgrade costs fall on local Elexicon and Hydro One Class B ratepayers?"[cite: 1, 2]

2. ACOUSTIC VIBRATIONS & LOW-FREQUENCY NOISE (DBC)

- **Tonal Penalty Incurred:** HGC Engineering's September 9, 2026 Environmental Noise Assessment models 13 outdoor Aermec TBA3350F chillers, 6 diesel generators (107 dBA each), and 6 step-up transformers (83 dBA each)[cite: 1]. Table A1 formally identifies every chiller and transformer as **"T" (Tonal)**, triggering a mandatory +5 dBA penalty that raises each chiller baseline to 94 dBA[cite: 1].
- **The 125 Hz & Infrasound Blindspot:** Page 37 of FirstBlock's filing specifies an unweighted sound power level of **100.3 dB at 125 Hz**[cite: 1]. Standard dBA weighting filters this down by –16.1 dB to 84.2 dBA, ignoring the wall-vibrating low-frequency pressure wave[cite: 1]. The filing completely omits the 31.5 Hz and 63 Hz octave bands[cite: 1].

- **Ecological & Residential Impact:** The resulting 65–72 dBC valley reverberation disrupts resident sleep architecture and directly interferes with the **100–500 Hz** substrate vibrational frequencies used by honeybees (*Apis mellifera*) for waggle-dance communication across Murray and Sidney Ward orchards[cite: 2, 4].
- **Wiki & Archive Evidence:** [\[\[raw/articles/04_acoustics_dba_dbc.md\]\]](#) | [\[\[raw/articles/datacenterally-com-blog-why-dba-is-not-the-whole-sound-story.md\]\]](#) | [\[\[raw/articles/loudoun-gov-6405-noise-air-quality-concerns.md\]\]](#) [cite: 1, 4]

DIRECT QUESTION FOR PROPONENTS:

"Why does HGC Engineering's study report compliance exclusively in dBA when your chiller manufacturer data on Page 37 reveals 100.3 dB of unweighted energy at 125 Hz, and will FirstBlock commit to a binding site-plan condition capping boundary noise at 55 dBC with continuous fenceline acoustic monitoring?"[cite: 1, 2]

3. FLOODPLAIN MAPPING & VERTICAL DATUM DISCREPANCY

- **Attack on Conservation Authority Mapping:** On Pages 31–46 of FirstBlock's October 2, 2026 filing, the proponents stamped **"FAULTY / WRONG DATA"** across Lower Trent Conservation's (LTC) February 2024 Regulatory Floodplain Map (Sheet 112)[cite: 1, 4].
- **Unconverted Survey Elevations:** On Page 42 of FirstBlock's comparison drawing, the footnote acknowledges:

"spot elevations from the IBW Surveyors topographic plan (May 29, 2026, datum CGVD28:78)... Regulatory flood elevations from the KGS Group / Lower Trent Conservation... (Feb. 2024, sheet 112, datum CGVD2013)... Clearances are before datum conversion."[cite: 1, 4]
- **Survey Defect:** Evaluating elevations across unconverted vertical reference systems (CGVD28:78 vs. CGVD2013) invalidates flood clearance calculations for outdoor generator yards and electrical equipment located adjacent to the Trent River corridor[cite: 1].
- **Wiki & Archive Evidence:** [\[\[concepts/floodplain-mapping.md\]\]](#) | [\[\[raw/articles/floodplain-mapping-cgvd28-cgvd2013.md\]\]](#) [cite: 1, 4]
- **Verification Status:** Verified per FirstBlock filing (Oct 2, 2026) Page 42 footnote CGVD28:78 (IBW Surveyors vs LTC Sheet 112 CGVD2013 'Clearances are before datum conversion') and Page 31 'FAULTY / WRONG DATA' stamp[cite: 1, 4].

DIRECT QUESTION FOR PROPONENTS:

"Your Page 42 drawing accuses Lower Trent Conservation of using 'Wrong Data,' yet your own footnote admits you compared CGVD28 survey elevations against CGVD2013 flood elevations 'before datum conversion.' What are the verified clearances once both datasets are harmonized, and has Lower Trent Conservation issued a formal Section 28 permit approving bulk diesel and chemical storage inside this regulated corridor?"[cite: 1, 2]

4. DRINKING WATER INTAKES & CHEMICAL SPILL RISKS

- **Direct Downstream Threat:** The facility sits immediately upstream of the **Trenton Water Treatment Plant (WTP) intake** (above Dam No. 1, serving ~7,000 ML/year across 8,800 connections) and the **Bayside WTP intake**, placing the site directly inside **Intake Protection Zones IPZ-2** (2-hour travel time) and **IPZ-3** under the *Ontario Clean Water Act, 2006*[cite: 1, 2].
- **Propylene Glycol Biological Oxygen Demand:** Closed-loop cooling chillers hold tens of thousands of liters of propylene glycol, which exerts a Biochemical Oxygen Demand of **1.68 kg O₂ per liter**[cite: 1, 2]. An uncontrolled rupture into the river rapidly strips dissolved oxygen below 2.0 mg/L, suffocating aquatic life and entering the **Bay of Quinte Area of Concern (AOC)**[cite: 1, 2].
- **Bulk Diesel Fuel Hazards:** Six 107 dBA standby diesel generators require tens of thousands of liters of bulk on-site hydrocarbon storage[cite: 1, 2]. Precedents from data center valve failures (e.g., Secaucus, NJ, Sept 2026; 5,000+ gallon spill) demonstrate high contamination risks to connected municipal water supplies[cite: 1].
- **Wiki & Archive Evidence:** [\[\[concepts/trent-river-drinking-water-intakes.md\]\]](#) | [\[\[raw/articles/trent_river_intakes_and_ipz_source.md\]\]](#) [cite: 1, 4]

DIRECT QUESTION FOR PROPONENTS:

"What is the exact combined volume of propylene glycol and diesel fuel to be stored on the banks of the Trent River, and will FirstBlock engineer 110% capacity impermeable secondary containment vaults with automated pneumatic shut-off valves tied to the

Trenton WTP intake alarm system?"[cite: 1, 2]

5. ZONING LOOPHOLE & MORATORIUM EXEMPTION TARGET

- **Administrative Site Plan Bypass:** FirstBlock relies on a June 12, 2026 email from Planning Director Brian Jardine stating that 869/920 Trenton-Frankford Road is zoned **GM (General Industrial)** and permitted for data center use, bypassing Section 34 public rezoning hearings under the *Planning Act*[cite: 1, 4].
- **Lobbying Against By-law 26-131:** FirstBlock's October 2 submission specifically requests Council enact **Sections 4A–4C** to exempt the Sonoco property from Quinte West's 1-year Interim Control By-law 26-131 (data center study moratorium) at the **October 7, 2026 Council meeting**[cite: 1, 4].
- **Contradictory Heat Export Data:** Proponents advertise site sustainability through vertical farming and aquaculture, yet technical figures conflict: Mayor Jim Harrison's December 4, 2025 support letter cites a **55–65°C** liquid-cooling loop, while MP Joël Lightbound's January 28, 2025 letter cites an **80°C** dual-loop heat exchanger system[cite: 1].
- **Wiki & Archive Evidence:** [\[\[raw/articles/zoning-loophole-gm-zone.md\]\]](#) [cite: 1, 4]
- **Verification Status:** Verified via Brian Jardine June 12, 2026 email (GM zone site plan control) and Section 4A-4C exemption lobby from Interim Control By-law 26-131 (Oct 7, 2026 Council vote)[cite: 1, 4].

DIRECT QUESTION FOR PROPONENTS:

"If FirstBlock's design meets all provincial and municipal environmental and acoustic standards, why are you demanding a custom exemption from Quinte West's 1-year Interim Control By-law 26-131 at tomorrow's Council meeting rather than participating in the city's comprehensive land-use study?"[cite: 1]

6. MUNICIPAL TAX DEFICIT & EMPLOYMENT REALITY

- **The Machinery Tax Exemption (MPAC):** Under **Section 3(1), paragraph 17 of the Ontario Assessment Act (R.S.O. 1990, c. A.31)**, all machinery and equipment used for processing (servers, GPU clusters, high-voltage transformers, switchgear, and UPS systems) is 100% exempt from municipal property taxes[cite: 2, 4].
- **The 'Shell vs. Contents' Deficit:** Over **80% to 90% of the capital expenditure** in a data center is concentrated in tax-exempt compute and power equipment[cite: 2]. Quinte West only collects industrial property taxes on the depreciated physical mill shell and raw land[cite: 2].
- **Automation vs. Job Promises:** Modern data centers operate virtually unstaffed, yielding an industry average of **0.5 to 2.0 permanent jobs per MW** (10 to 25 permanent employees for the entire 18–33 MW facility), refuting claims of broad local industrial employment[cite: 1, 2, 4].
- **Wiki & Archive Evidence:** [\[\[raw/articles/municipal-tax-deficit-ontario-assessment.md\]\]](#) [cite: 1, 4]
- **Verification Status:** Verified under Ontario Assessment Act R.S.O. 1990 c. A.31 s. 3(1) para 17 machinery exemption and 0.5-2.0 permanent jobs/MW industry baseline[cite: 2, 4].

DIRECT QUESTION FOR PROPONENTS:

"Under the Assessment Act machinery exemption, what is the exact net property tax revenue Quinte West will collect annually above what the mill shell already pays, and how many permanent full-time jobs are contractually guaranteed to work directly inside the data center if the vertical farm fails to launch?"[cite: 2]

RESIDENT ACTION STRATEGY FOR THE FLOOR

1. **Demand Written Commitments:** Request that all claims regarding grid capacity, decibel ceilings, and flood datums be submitted in writing to the City Clerk before tomorrow's Council vote[cite: 1].
2. **Insist on ICBL Inclusion:** Urge Council to vote **NO** on the Section 4A–4C custom exemption to Interim Control By-law 26-131 on October 7, ensuring the Sonoco site remains subject to the city's comprehensive 1-year data center study[cite: 1].